

Dongwook Kwon

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Experience

Suresoft Technologies

Seongnam, Gyeonggi, Korea

AI ENGINEER, INTELLIGENT AGENT TECHNOLOGY TEAM

Dec. 2025 – Present

- Developing an AI validation system utilizing AI agent-based architectures.
- Designing and implementing systems to verify and evaluate AI models through autonomous and collaborative AI agents.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University

Seoul, Korea

RESEARCH ASSISTANT

Aug. 2021 – Dec. 2025

- Conducted research on deep learning algorithms, focusing on anomaly detection in time-series data, including Electrocardiogram (ECG) and Cyber-Physical Systems (CPS).
- Participated in the development of an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

LG Electronics Canada

Toronto, ON, Canada

RESEARCH PROJECT

Jan. 2025 – Jul. 2025

- Participating in the development of an assessment framework for agentic AI systems utilizing LLMs to evaluate performance.

Qualcomm Institute, University of California San Diego

San Diego, CA, USA

AI DEVELOPMENT PROJECT, SUMMER PROGRAM

Jul. 2022 – Aug. 2022

- Participated in the Personality-Based Drug Addiction Prediction System project.
- Attended seminars on deep learning and machine learning algorithms.

Education

Kwangwoon University

Seoul, Korea

MASTER OF SCIENCE IN COMPUTER ENGINEERING

Mar. 2024 – Feb. 2026

- GPA: 4.33/4.5 (Percentage: 98.1%)

University of Toronto

Toronto, ON, Canada

GRADUATE RESEARCH PROGRAM IN MECHANICAL AND INDUSTRIAL ENGINEERING

Jan. 2025 – Jul. 2025

- GPA: 3.68/4.0

Kwangwoon University

Seoul, Korea

BACHELOR OF SCIENCE IN ELECTRONICS AND COMMUNICATIONS ENGINEERING

Mar. 2019 – Feb. 2024

- GPA: 3.41/4.5 (Percentage: 88.1%)

Honors & Awards

MAJOR AWARDS

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego

United States

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project - Minister of Oceans and Fisheries

South Korea

ACADEMIC RECOGNITION

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers

South Korea

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024)

South Korea

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering

South Korea

COMPETITIONS

Dec. 2025 **Silver Award**, 23rd Embedded Software Competition (KIISE)

South Korea

Dec. 2022 **Bronze Award**, Hanium ICT Mentoring Competition - The Federation of Korean Information Industries (FKII)

South Korea

Sep. 2022 **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University

South Korea

Oct. 2021 **Grand Prize**, MY (Multi-Y) Capstone Design Competition — Kwangwoon University

South Korea

Projects

Agentic AI Evaluation Framework

LG ELECTRONICS CANADA

- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

Toronto, ON, Canada

Jan. 2025 – Present

AI-based Intrusion Detection and Attack Packet Crafting Technologies for APR1400

SEJONG UNIVERSITY

- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Seoul, Korea

Jul. 2022 – Present

Quadruped Robot Using Vision SLAM with a Depth Camera

MINISTRY OF SCIENCE AND ICT

- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Seoul, Korea

Apr. 2022 – Nov. 2022

Personality-Based Drug Addiction Prediction System

QUALCOMM INSTITUTE, UNIVERSITY OF CALIFORNIA SAN DIEGO

- Developed a personality-based drug addiction prediction system using a machine learning model.

San Diego, CA, USA

Jul. 2022 – Aug. 2022

Development of Active Kill-Switch and Biomarker-Based Defense System for Life-Threatening IoT Medical Devices

KOOKMIN UNIVERSITY

- Developed a security system for heart implant devices using an anomaly detection algorithm.

Seoul, Korea

Sep. 2021 – Dec. 2022

Development of a Smart Logistics System Using Computer Vision

MINISTRY OF OCEANS AND FISHERIES

- Developed a logistics detection system incorporating location tracking and stack layer identification for containers.

Seoul, Korea

Apr. 2021 – Nov. 2021

Smart Kitchen IoT System Utilizing Computer Vision and Deep Learning

KWANGWOON UNIVERSITY

- Developed a cloud IoT platform featuring a system for classifying groceries in the fridge and an OCR receipt program.

Seoul, Korea

May 2021 – Oct. 2021

Publications

Journal Articles

- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.
- **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

Conference Papers

- R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) – Industry Track (Poster)*, 2026.
[†]Equal contribution
- J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025.
[†]Equal contribution
- **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024.
Best Poster Award – Silver Winner
- **D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023.
Outstanding Paper Award

Patents

- **Issued Patent**, "Method for Implantable Medical Device to Detect Anomaly in Real Time", Patent No. KR 10-2961984, Issued on April 30, 2026.
- **Issued Patent**, "GPT API-Based Network Anomaly Detection", Patent No. KR 10-2848814, Issued on August 18, 2025.

Leadership Experience

- **President**, Kwangwoon International Student Association - Lead intercultural programs and student engagement initiatives.
- **Director of Filming and Editing**, Kwangwoon Broadcasting Center (KWBC) – Oversaw media production and content delivery.
- **Class Representative**, Dept. of Electronics and Communications Engineering – Represented student interests in academic affairs.
- **Executive Member**, Amateur Radio Club – Organized radio workshops and technical exhibitions.